

$$\iint_D f(x, y) d\sigma = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta \sigma_i f(x_i, y_i)$$

(1) 值与积分区域与被积函数有关, 与变量表示无关.

(2) 二重积分存在 $\Rightarrow f(x, y)$ 在有界闭区域 D 上有界

(3) 改变有限个点值不变

☆ ③ $f(x, y)$ 在 D 区域上 $\equiv 1$, $\iint_D f(x, y) d\sigma = \iint_D 1 d\sigma = \underline{S_D}$

④ $f(x, y) \leq g(x, y)$, $\iint_D f(x, y) d\sigma \leq \iint_D g(x, y) d\sigma$; $\downarrow$ D 区域的面积
(保号性)

特别的, 若 $f(x, y), g(x, y)$ 连续且 $f(x, y) \neq g(x, y)$, 则 ~~不等于~~ $\iint_D f(x, y) d\sigma < \iint_D g(x, y) d\sigma$

⑤ 估值定理 $f(x, y)$ 在 D 上可积且连续 $m \leq f(x, y) \leq M$.

则 $m \cdot S_D \leq \iint_D f(x, y) d\sigma \leq M \cdot S_D$

⑥ 中值定理 $f(x, y), g(x, y)$ 在 D 上连续且 $g(x, y)$ 在 D 上不变号 D 上至少

有一点 (ξ, η) , 使 $\iint_D f(x, y) g(x, y) d\sigma = f(\xi, \eta) \iint_D g(x, y) d\sigma$

变式: $g(x, y) \equiv 1$, $\iint_D f(x, y) d\sigma = S_D \cdot f(\xi, \eta)$

(积分中值定理)

计算 ① 直角坐标系: $d\sigma = dx dy$

形成区域: 1. x型区域

$$\begin{cases} a \leq x \leq b \\ y_1(x) \leq y \leq y_2(x) \end{cases} \quad (\text{定 } y \text{ 时上下限可包含 } x)$$

先定 x , 再平行 y 轴穿针, 沿 y 轴正方向穿过积分区域

$$\iint_D f(x, y) d\sigma = \int_a^b dx \int_{y_1(x)}^{y_2(x)} f(x, y) dy$$

2. y型区域

先定 y , 再平行 x 轴穿针, 沿 x 轴正方向穿过积分区域

★ 不管什么形区域, 穿针交点不可超过两个, 超过则应拆分

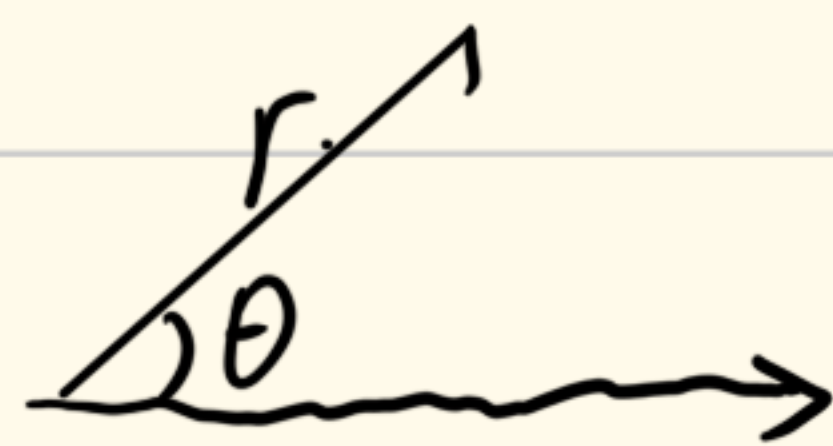
考点: 交换积分次序

① 根据积分区域表达难易

② 根据被积函数是否便于求值

②极坐标

直角坐标与极坐标互化:



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \end{cases}$$

$$dx dy = r dr d\theta$$

$$\iint_D f(x, y) dx dy = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$

计算: (确定积分限先定 θ ,再定 r)

$$x^2 + y^2 = 4 \ (x, y \geq 0), \ y \geq \sqrt{2}, \ x = 0$$



$$\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

$$\frac{\sqrt{2}}{\sin \theta} \leq r \leq \frac{2}{\sqrt{x^2 + y^2} = 4}$$

(技巧, 把 x, y 代入, 先写直角坐标系下范围)

$$\iint_D f(x) d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_{\frac{\sqrt{2}}{\sin \theta}}^2 f(r \cos \theta, r \sin \theta) r dr$$

使用: ① 积分域与圆有关

② 被积函数有 $f(x^2+y^2)$ 类似形式

e.g. $\iint_D e^{x^2+y^2} dx dy$, D 由 x 轴与 $y = \sqrt{1-x^2}$ 围成



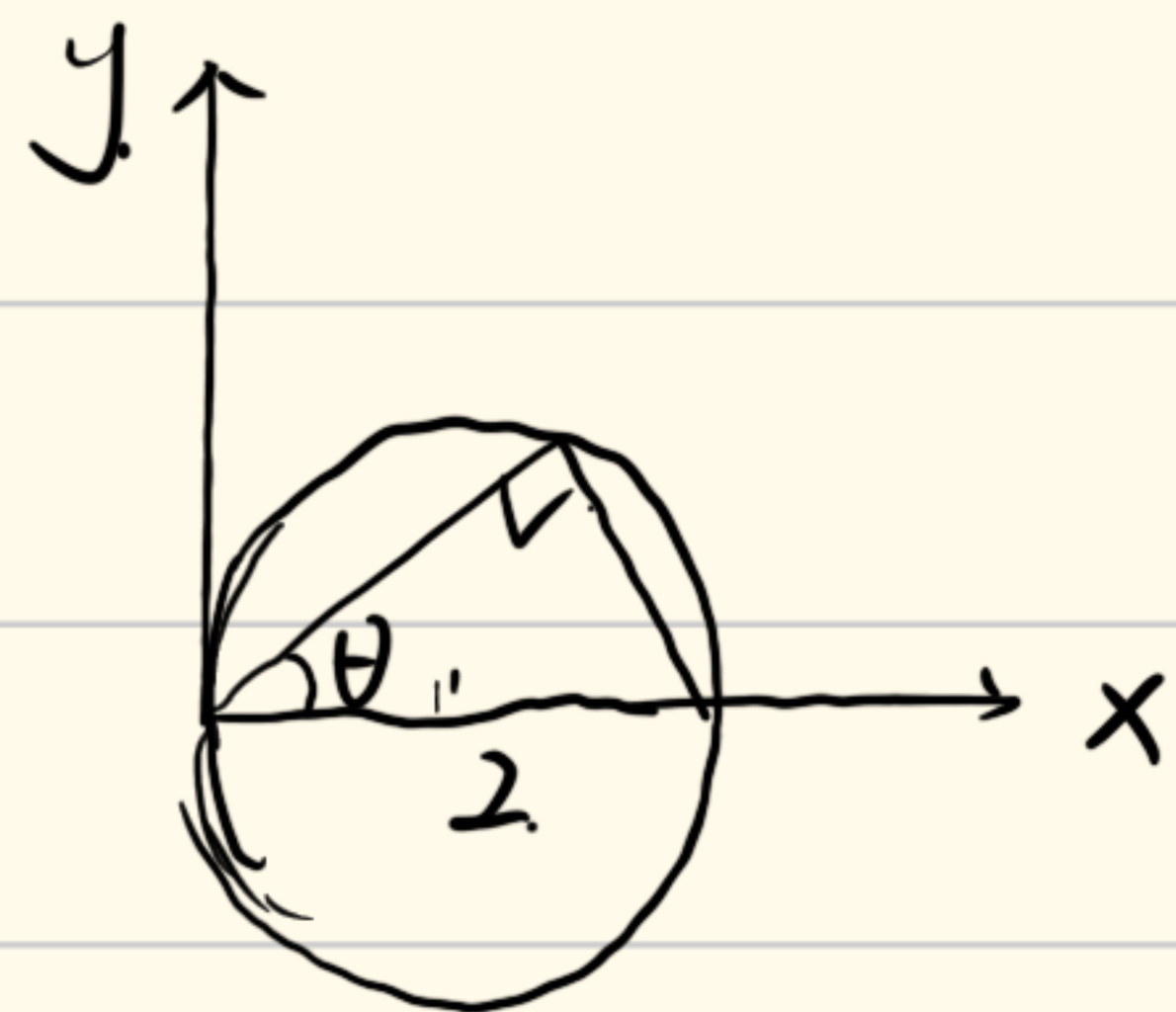
$$0 \leq \theta \leq \pi$$

$$0 \leq r \leq 1$$

$$\begin{aligned} & \iint_D e^{x^2+y^2} r dr d\theta \\ &= \int_0^\pi d\theta \int_0^1 e^{r^2} r dr \\ &= \pi \cdot \frac{1}{2} e^{r^2} \Big|_0^1 \\ &= \pi \cdot \left(\frac{1}{2} e - \frac{1}{2} \right) \end{aligned}$$

tips: 若二元函数 $u(x,y) = f(x) \cdot g(y)$, 且积分上下限为常数, 则二重积分可拆为两定积分之积

e.g. 2. $I = \iint_D \sqrt{x^2+y^2} dx dy$, $D: x^2+y^2 = 2x$ 围成



$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$0 \leq r \leq 2 \cos \theta$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2 \cos \theta} r^2 dr.$$

$$= \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 \theta d\theta$$

$$= \frac{16}{3} \int_0^{\frac{\pi}{2}} \cos^3 \theta d\theta$$

$$= \frac{16}{3} \int_0^{\frac{\pi}{2}} (1 - \sin^2 \theta) d\sin \theta$$

$$= \frac{16}{3} \cdot \left(\sin \theta - \frac{1}{3} \sin^3 \theta \right) \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{32}{9}$$

点火公式

$$\textcircled{1} \int_0^{\frac{\pi}{2}} \sin^n x = \int_0^{\frac{\pi}{2}} \cos^n x dx = \begin{cases} \frac{(n-1)!!}{n!!}, n=2k-1 \\ \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2}, n=2k \end{cases}$$

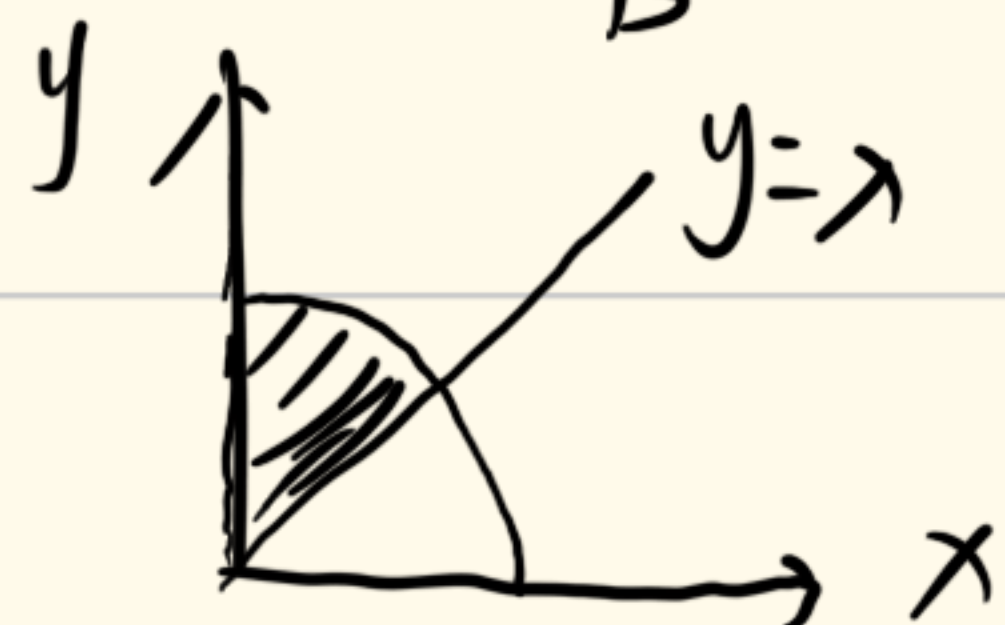
$$\textcircled{2} \int_{-\pi}^{\pi} \sin^n x dx = 2 \int_0^{\frac{\pi}{2}} \sin^n x dx = 2I_n$$

$$\int_0^{\pi} \cos^n x dx = \begin{cases} 0, n=2k+1 \\ 2 \int_0^{\frac{\pi}{2}} \cos^n x dx, n=2k \end{cases}$$

$$\textcircled{3} \int_0^{2\pi} \sin^n x = \int_0^{2\pi} \cos^n x dx = \begin{cases} 0, n=2k+1 \\ 4I_n, n \text{ 为偶} \end{cases}$$

$$\textcircled{4} \int_0^{2\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

eg. 3. $I = \iint_D y dx dy$, $D: y = \sqrt{1-x^2}, y = x, x = 0$



$$\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

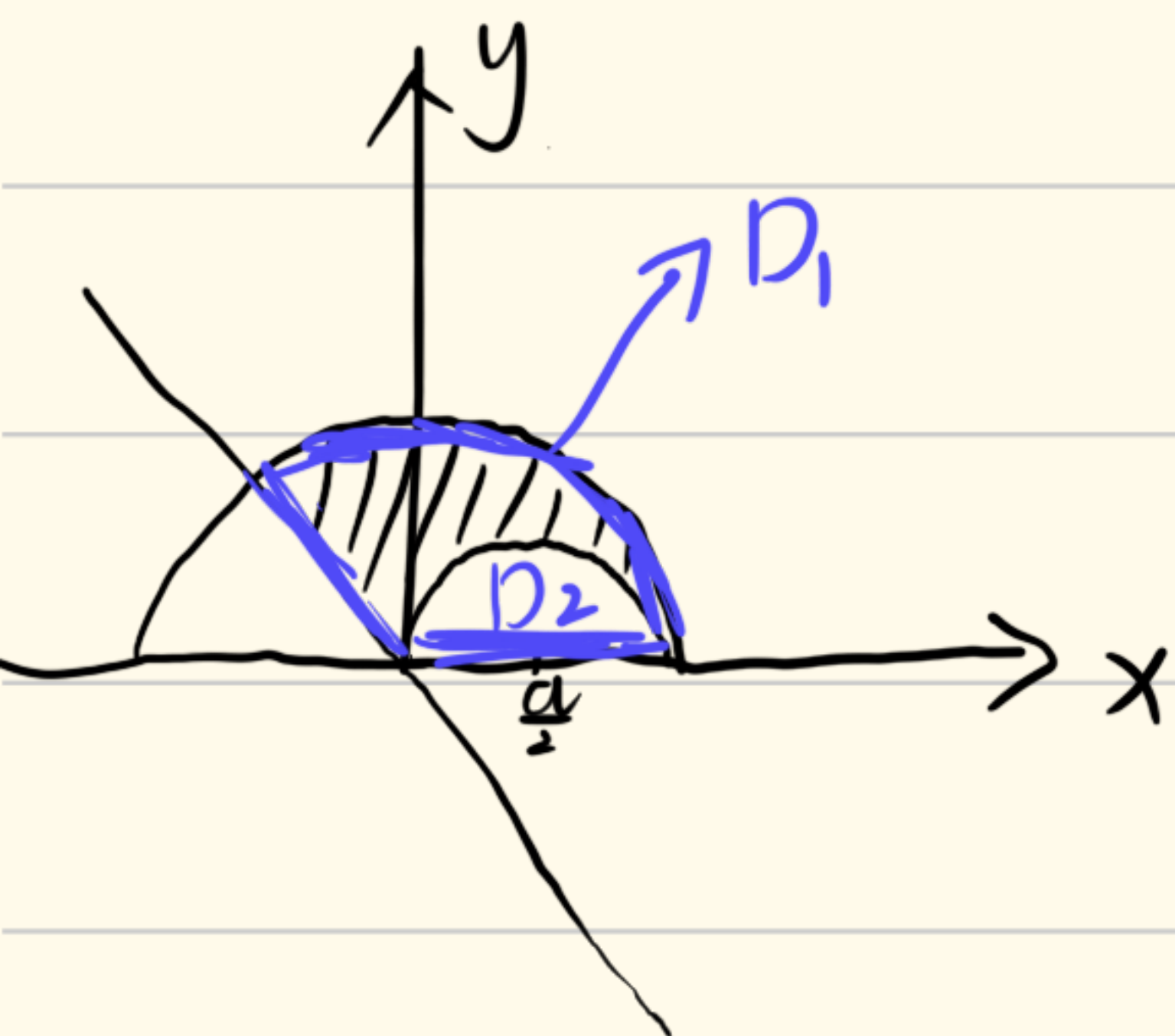
$$0 \leq r \leq 1$$

区间再现公式: 若 $f(x)$ 连续, 则

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$\begin{aligned}
 & \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \sin\theta \, dr. \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin\theta \, d\theta \cdot \int_0^1 r^2 \, dr. \\
 &= -\cos\theta \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot \frac{1}{3} r^3 \Big|_0^1 \\
 &= \frac{\sqrt{2}}{6}
 \end{aligned}$$

eg 4 $I = \iint_D \frac{dx dy}{\sqrt{x^2+y^2}}$, $D: y = \sqrt{a^2-x^2}, y = \sqrt{ax-x^2}, y = -x \ (a>0)$



$$0 \leq \theta \leq \frac{3\pi}{4}$$

看作 $\underline{D_1} - \underline{D_2}$
 ↓ ↓
 大扇形 小偏圆

二重积分的对称性

1. 奇偶对称性

$$\text{积分区域 } D \text{ 关于 } \begin{cases} x \text{ 轴对称} \\ y \text{ 轴对称} \end{cases} \begin{cases} f(x, y) = -f(x, -y) \Rightarrow \iint_D f(x, y) d\sigma = 0 \\ f(x, y) = f(x, -y) \Rightarrow 2 \iint_{D_1} f(x, y) d\sigma, D_1: x > 0 \end{cases}$$

$$\begin{cases} f(x, y) = -f(-x, y) \Rightarrow \iint_D f(x, y) d\sigma = 0 \\ f(x, y) = f(-x, y) \Rightarrow 2 \iint_{D_1} f(x, y) d\sigma \end{cases}$$

$$D \text{ 关于原点对称} \quad f(x, y) = -f(-x, -y) \quad I = 0$$

$$f(x, y) = f(-x, -y) \quad I = 2 \iint_{D_3} f(x, y) d\sigma, D_3: D \text{ 的一半区域}$$

若 D 关于某一直线对称, 在对称点处被积函数值互为相反数, $I = 0$

2. 轮换对称性.

$$1) \text{关于 } y=x \text{ 对称, } \iint_D f(x,y) d\sigma = \iint_D f(y,x) d\sigma = \frac{1}{2} \iint_D [f(x,y) + f(y,x)] d\sigma$$



积分区域表达式 x, y 互换, 形式不变

★: 对于题目先在直角坐标系下应用对称性简化被积函数, 再进行求解

$$\begin{aligned} 4 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos^4 \theta d\theta &= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta = 8 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta \\ &= 8 \cdot \frac{3 \times 1}{4 \times 2} \times \frac{\pi}{2} = \frac{3\pi}{2} \end{aligned}$$

三重积分

1. 性质 $f(x, y, z) \equiv 1$, $\iiint_{\Omega} f(x, y, z) dV = V_{\Omega}$

(估值定理) 连续 $m \leq f(x, y, z) \leq M$

$$m V_{\Omega} \leq \iiint_{\Omega} f(x, y, z) dV \leq M \cdot V_{\Omega}$$

2. 计算

① 先一后二(穿针法) 先定积分, 后二重积分

e.g. 平行于z轴穿针

step 1 画出积分区域 Ω 在 xy 面的投影 D_{xy} , 从 D_{xy} 内任一点沿 z 轴正方向穿过 Ω 先穿过的面 $\underbrace{z_1(x, y)}_{\downarrow \text{下限}}$, 后穿过的面: $\underbrace{z_2(x, y)}_{\uparrow \text{上限}}$

step 2: 在 D_{xy} 区域进行二重积分

$$\int_a^b dx \int_{y_1(x)}^{y_2(x)} dy \int_{z_1(x, y)}^{z_2(x, y)} f(x, y, z) dz$$

总结: 平行于某坐标轴穿针, 投影到另外两个变量所在平面

② 先二后一 (截面法)

假定平行 xOy 平面进行截面

step 1: 确定 z 的取值范围

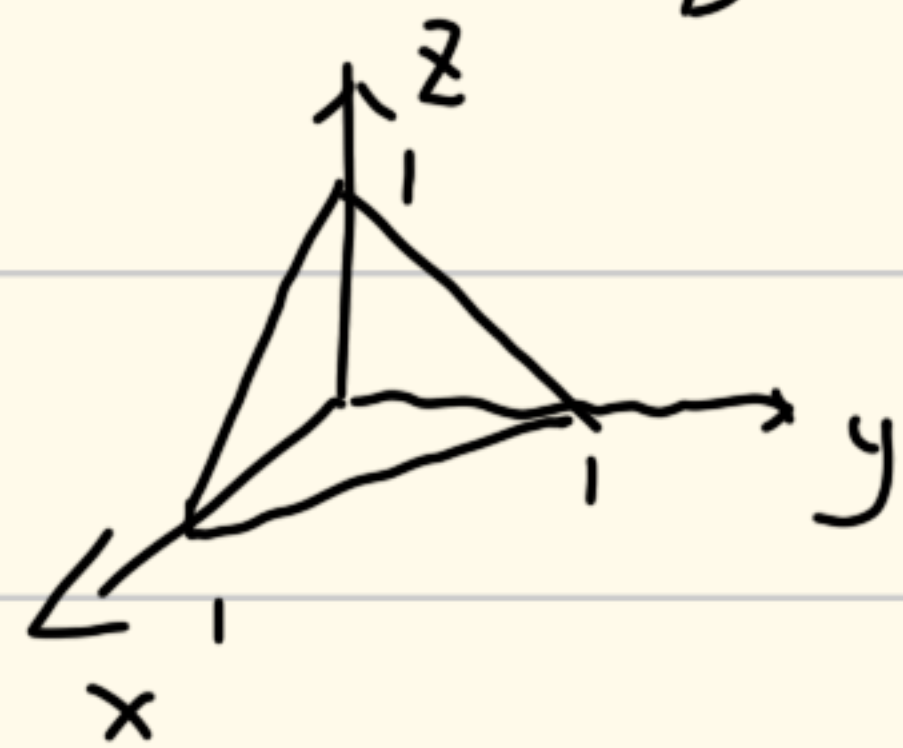
step 2: 固定 z 确定每个截面的表达式 $\int_{c_1}^{c_2} dz \iint_{D_z} f(x, y, z) dx dy$

适用范围: ① 被积函数是单变量函数 (变量是什么就垂直于此坐标轴截面)

e.g. $f(z)$ $\int_{c_1}^{c_2} dz \underbrace{\iint_{D_z} f(z) dx dy}_{\Delta}$

② 对应截面面积易求

e.g. $I = \iiint_D y dv$, Ω 由三坐标面及 $x+y+z=1$ 围成



法1: 穿针法

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1-x$$

$$0 \leq z \leq 1-x-y$$

$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} y dz$$

$$= \int_0^1 dx \int_0^{1-x} y(1-x-y) dy$$

$$= \int_0^1 \left(-\frac{1}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{6} \right) dx$$

$$= \frac{1}{24}$$

$$y - xy - y^2$$

$$\frac{1}{2}y^2 - \frac{1}{2}xy^2 - \frac{1}{3}y^3$$

$$\frac{1}{2} \cdot (1-2x+x^2) \cdot (1-x) - \frac{1}{3} \cdot \frac{1}{(1-3x+3x^2-3x^3)}$$

2: 截面法

含哪个变量用哪个截

$$\int_0^1 dy \iint_{D_y} y dx dz = \int_0^1 y \left[\int_{D_y} dx dz \right] dy$$

$$= \int_0^1 y \cdot \frac{1}{2} (1-y)^2 dy = \frac{1}{24}$$

e.g. 2 $\iiint z dx dy dz$, $\Omega: z = x^2 + y^2$ 与 $z=1$



1. 穿针法

$$-1 \leq x \leq 1$$

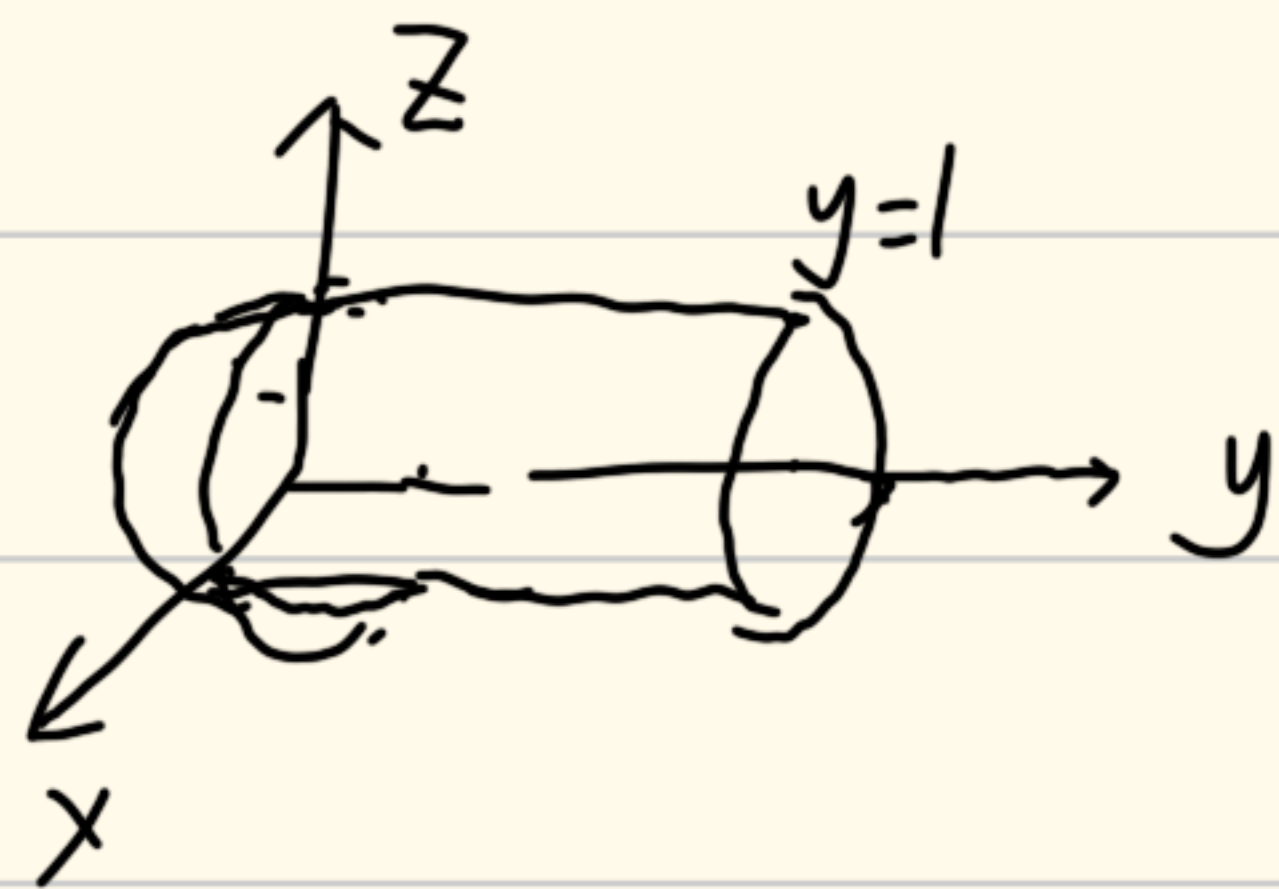
$$-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}$$

$$x^2 + y^2 \leq z \leq 1$$

2. 截面法

$$\begin{aligned} \int_0^1 dz \iint_{D_z} z dx dy &= \int_0^1 z \cdot \pi \cdot z \cdot dz \\ &= \int_0^1 \pi z^2 dz = \frac{1}{3} \pi \end{aligned}$$

e.g. 3. $\iiint y \sqrt{1-x^2} dV$, $y = -\sqrt{1-x^2-z^2}$, $x^2+z^2=1$, $y=1$.



$$-1 \leq x \leq 1$$

$$-\sqrt{1-x^2} \leq z \leq \sqrt{1-x^2}$$

$$-\sqrt{1-x^2-z^2} \leq y \leq 1$$

$$x = r \cos \theta, z = r \sin \theta, y = y$$

$$\int_0^{2\pi} d\theta \int_0^1 dr \int_{-\sqrt{1-r^2}}^1 y r^2 \sin \theta dy$$

$$\left. \frac{1}{2} y^2 r \sin \theta \right|_{-\sqrt{1-r^2}}^1$$

$$\int_0^1 \left(\frac{1}{2} r^2 \sin \theta - \frac{1}{2} (1-r^2) r^2 \sin \theta \right) dr$$

$$\left. \frac{1}{4} r^2 \sin \theta \right|_0^1 - \frac{1}{2} (r \sin \theta - r^3 \sin \theta)$$

$$\frac{1}{4} \sin \theta - \frac{1}{8} \sin \theta$$

$$\left. \frac{1}{8} \sin \theta \right|_0^{2\pi}$$

$$\frac{1}{2} r^3 \sin \theta - \frac{1}{2} r \sin \theta$$

$$\left. \frac{1}{8} r^4 \sin \theta - \frac{1}{4} r^2 \sin \theta \right|_0^1$$

③柱坐标

= 极坐标 + z 分量

$$x, y, z \rightarrow r, \theta, z.$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \Rightarrow r^2 = x^2 + y^2.$$

计算:

相当于先一后二的穿针

先穿针后利用极坐标在投影面上二重积分

$$\iiint_V f(x, y, z) dV = \int_a^b d\theta \int_{r_1(\theta)}^{r_2(\theta)} dr \int_{z_1(r, \theta)}^{z_2(r, \theta)} r \cdot f(r \cos \theta, r \sin \theta, z) dz$$

适用范围: 投影区域为圆的一部分, 被积函数中有 $f(x^2 + y^2)$ 形式

e.g.1 $\iiint_{\Omega} z \, dx \, dy \, dz, \quad z = x^2 + y^2, \quad z = 1$

$$\int_0^{2\pi} d\theta \int_0^1 dr \int_{r^2}^1 r \cdot z \, dz = \frac{1}{2} \int_0^{2\pi} d\theta \int_0^1 r(1-r^4) \, dr.$$

$$= \frac{1}{2} \times \frac{1}{3} \times 2\pi.$$

$$= \frac{2\pi}{3}$$

$$\frac{1}{2} z^2 \Big|_{r^2}^1$$

$$\frac{1}{2} - \frac{1}{2} r^4$$

e.g.2 $\iiint_{\Omega} (x^2 + y^2) \cdot dx \, dy \, dz, \quad z = \sqrt{x^2 + y^2}, \quad z = a$

$$\int_0^{2\pi} d\theta \int_0^a dr \int_r^a r^3 \, dz$$

$$= 2\pi \int_0^a (a-r) \cdot r^3 \, dr.$$

$$= \frac{1}{10} a^5 \pi$$

$$r - \frac{1}{5} r^5.$$

$$r - r^5$$

$$\frac{1}{2} - \frac{1}{6}$$

$$ar^3 - r^4$$

$$\frac{1}{4} ar^4 - \frac{1}{5} r^5$$

$$\frac{1}{4} a^5 - \frac{1}{5} a^5$$

$$\iiint_{\Omega} \sqrt{x^2+y^2} \, dx \, dy \, dz$$

$$\Omega: x^2+y^2+(z-1)^2 \leq 1$$

$$(z-1)^2 = 1-r^2 \quad z-1 = \sqrt{1-r^2}$$

~~0~~

$$\int_0^{2\pi} d\theta \int_0^1 dr \int_{1-\sqrt{1-r^2}}^{1+\sqrt{1-r^2}} r^2 \, dz$$

$$24 - \frac{63}{6}$$

$$42 \int_0^1 \sqrt{1-r^2} r^2 \, dr = 42 \cdot \frac{2}{16} = \frac{21}{4}$$

$$\int_0^{\frac{\pi}{2}} r^2 \sin^2 \theta \, d\theta$$

$$\int_0^{\frac{\pi}{2}} (\sin^2 \theta - \sin^4 \theta) \, d\theta$$

$$\frac{1}{2} \cdot \frac{2}{2} - \frac{3}{8} \cdot \frac{2}{2}$$

$$16r - r^5$$

$$8r^2 - \frac{1}{6}r^6 \Big|_1^2$$

$$\frac{1}{2}r \cdot (16-r^4)$$

$$\iiint_{\Omega} z \, dx \, dy \, dz, \quad z = x^2+y^2, \quad z=1, \quad z=4$$



①补

②割

$$\begin{aligned} & \int_0^{2\pi} d\theta \int_1^2 dr \int_{r^2}^4 z r \, dz + \int_0^{2\pi} \int_0^1 dr \int_1^4 z r \, dz \\ &= 2\pi \cdot \int_1^2 \frac{1}{2} r (16-r^4) \, dr + 2\pi \cdot \frac{15}{2} \cdot \frac{1}{2} \\ &= \frac{27}{2}\pi + \frac{15}{2}\pi = 21\pi \end{aligned}$$

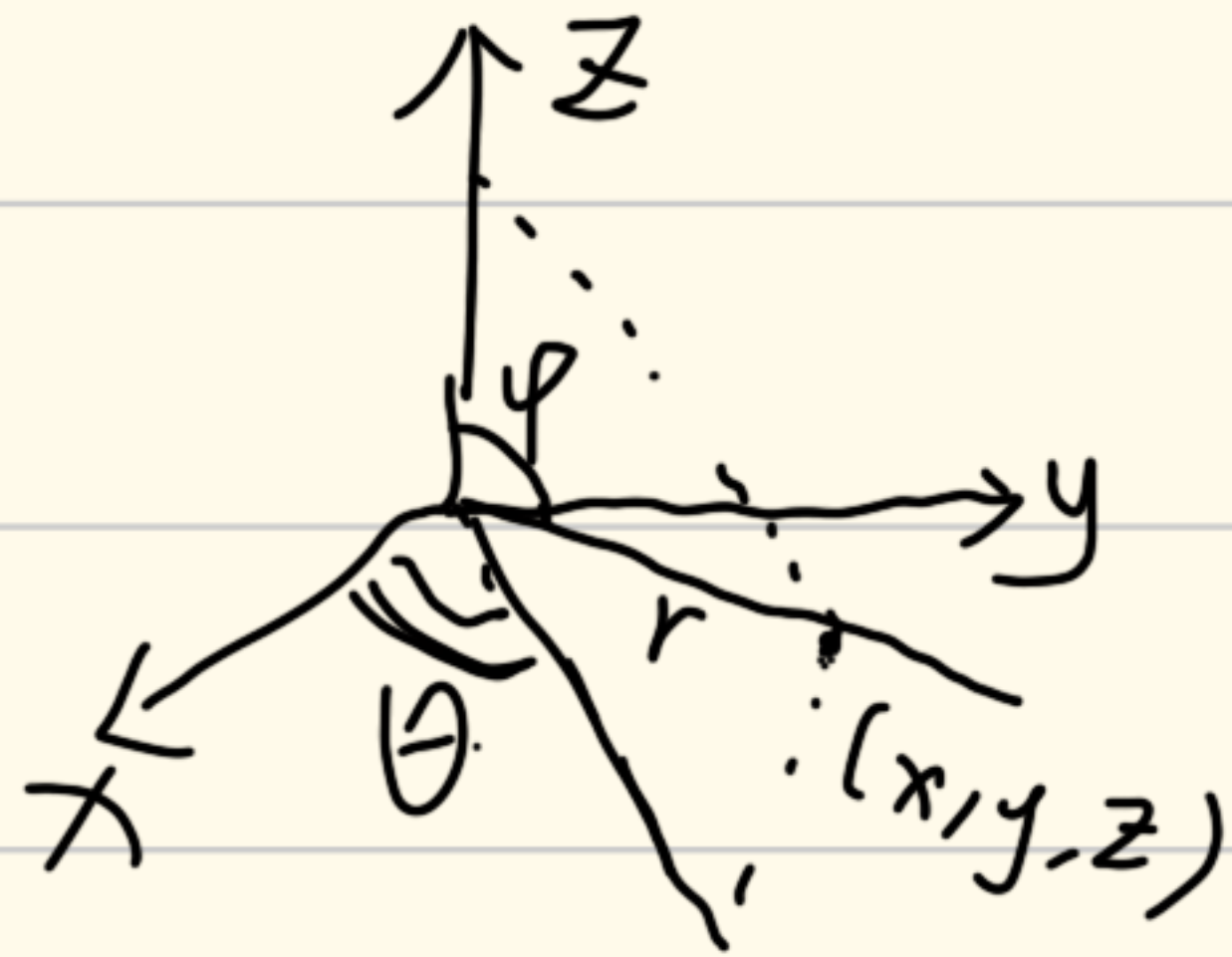
$$\frac{1}{2} \pi^2 (16-1)$$

截面法 $\int_1^4 dz \cdot \iint_{D_z} z \cdot dx dy = \int_1^4 z dz \iint_{D_z} 1 dx dy = \int_1^4 z \pi z^2 dz$

$$= 2\pi \left[\frac{1}{3} z^3 \right]_1^4 = \frac{2\pi}{3} (64 - 1)$$

④球坐标

1. 球坐标系 (由两个曲面和一个平面确定空间中一点的位置)



r : 以原点为球心, 半径为 r 的球面

θ : 以 z 轴为边缘的半平面

φ : 以原点为顶点, z 轴为中心轴, 半顶角为 φ 的锥面
($0 \leq \varphi \leq \pi$)

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$$

$$x^2 + y^2 + z^2 = r^2 \Rightarrow (x, y, z) \text{ 在球面上}$$

$$dV = r^2 \sin \varphi \, dr \, d\theta \, d\varphi$$

2. 计算: (先定 θ , 再定 φ , 再定 r .)

θ : 积分区域 Ω 投到 xOy 面上, 与 x 轴正方向的夹角

φ : 固定 θ 值与 z 轴的夹角 $\varphi_{\min} \rightarrow \varphi_{\max}$

r : 从原点发出射线, 穿过 Ω 区域, 穿入(下限) $r_1(\varphi, \theta)$, 穿出 $r_2(\varphi, \theta)$

3. 应用: 积分区域为球面/锥面 被积函数为 $f(x^2+y^2+z^2)$ 形式

e.g. $\iiint_{\Omega} (x^2+y^2) dV, \quad x \geq 0, y \geq 0, z \geq 0, x^2+y^2+z^2 \leq 4$ $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$\int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^2 r^4 \sin^3 \varphi \, dr$$

$$= -\frac{\pi}{2} \cdot \frac{2}{3} \times \frac{1}{5} \times 32 = \frac{32}{15} \pi$$

Δ 柱坐标

$$\int_0^{\frac{\pi}{2}} d\theta \int_0^2 dr \cdot \int_0^{\sqrt{4-r^2}} r^3 \, dz = \int_0^{\frac{\pi}{2}} d\theta \int_0^2 r^3 \sqrt{4-r^2} \, dr$$

$$= \frac{20}{2} \cdot \underbrace{\int_0^2 r^3 \sqrt{4-r^2} dr}_{r=2\sin t}$$

$$\int_0^{\frac{\pi}{2}} 8 \sin^3 t \cdot 2 \cos t \cdot 2 \cos t dt = \int_0^{\frac{\pi}{2}} 32 \sin^3 t \cdot \cos^2 t dt$$

$$= 32 \int_0^{\frac{\pi}{2}} \sin^3 t dt - 32 \int_0^{\frac{\pi}{2}} \sin^5 t dt$$

$$= 32 \cdot \left(\frac{2}{3} - \frac{8}{15} \right)$$

$$= \frac{64}{15}$$

eg. 2 $\iiint_V z dV, \quad z \leq \sqrt{x^2+y^2} \leq \sqrt{3}z, \quad 0 \leq z \leq 4$

$$z^2 \leq x^2+y^2 \leq 3z^2 \quad (\text{两个锥面中间的区域})$$

法一：直角坐标系：截面法

$$\int_0^4 dz \iint_D z dx dy$$

$$\frac{1}{4} z^4$$

$$= \int_0^4 z \cdot 2\pi \cdot 2z^2 dz = \cancel{\int_0^4 z} \cdot 2\pi \int_0^4 z^3 dz = 128\pi$$

法二：柱坐标系

割: $\int_0^{2\pi} d\theta \int_0^4 dr \int_{\frac{r}{\sqrt{3}}}^r z \cdot r dz = \frac{2\pi}{3} \int_0^4 r^3 \cdot dr = \frac{2}{3} \times 64\pi$

$\frac{1}{2} z^2$ $\frac{1}{2} r^2 - \frac{1}{2} \cdot \frac{r^2}{3}$

$\int_0^{2\pi} d\theta \int_4^{4\sqrt{3}} dr \int_{\frac{r}{\sqrt{3}}}^4 z \cdot r dz$ $\frac{1}{4} r^4$

$\frac{1}{2} z^2 \cdot r = \frac{1}{2} \cdot (16 - \frac{r^2}{3}) \cdot r = 8r - \frac{r^3}{6}$

太麻烦

$4r^2 - \frac{r^4}{24} \Big|_4^{4\sqrt{3}}$

$4r - \frac{r^2}{24} \Big|_{16}^{16 \times 3}$

$16 \times 3 \times 4 - \frac{16^2 \times 9}{24}$

$\frac{16}{32} = \frac{16}{192}$

补: $\int_0^{2\pi} d\theta \int_0^4 dr \int_r^4 z \cdot r dz$

$\frac{1}{2} z^2 \Big|_r^4 \cdot r = \int_0^4 (8 - \frac{1}{2} r^2) \cdot r dr$

$4r^2 - \frac{1}{8} r^4 \Big|_0^4$

$64 - \frac{4^3}{8} = 0$

$16 - 2 = 14$

64π

32π

$\int_0^{2\pi} d\theta \int_0^{4\sqrt{3}} dr \int_{\frac{r}{\sqrt{3}}}^4 z r dz$

$r \cdot \frac{1}{2} z^2 \Big|_{\frac{r}{\sqrt{3}}}^4 = \frac{1}{2} r \cdot (16 - \frac{r^2}{3})$

$8r - \frac{r^3}{6}$

$4r^2 - \frac{r^4}{24} \Big|_0^{4\sqrt{3}}$

$4r - \frac{r^2}{24} \Big|_0^{16 \times 3}$

$= 2 \times 96\pi = 192\pi$

法3: 球坐标系

$$\int_0^{2\pi} d\theta \int_{\frac{\pi}{4}}^{\frac{3}{4}} d\varphi \int_0^{\frac{4}{\cos\varphi}} r^3 \sin\varphi \cdot \cos\varphi dr. \quad 0 \leq r \leq \frac{4}{\cos\varphi} \quad 16 \times 12 - \frac{4^3 \times 9}{4 \times 3 \times 2} 32$$

$$= 2\pi \cdot \int_{\frac{\pi}{4}}^{\frac{3}{4}} \sin\varphi \cos\varphi d\varphi \int_0^{\frac{4}{\cos\varphi}} r^3 dr \Rightarrow \frac{1}{4} \times 8^4 = \frac{1}{4} \times 2^4 \times 4^3$$

$$\Rightarrow \sin\varphi d\sin\varphi$$

$$= 16 \times 64$$

$$\frac{1}{2} \sin^2\varphi. \quad \frac{1}{2} \left(\frac{3}{4} - \frac{1}{2} \right) = \frac{1}{8}$$

$$\frac{1}{4} r^4 = \frac{1}{4} \cdot \frac{4^4}{\cos^4\varphi}$$

$$= 2\pi \cdot \frac{16 \times 64}{8} = 16^2 \pi$$

$$= \frac{64}{\cos^4\varphi}$$

$$128\pi \int_{\frac{\pi}{4}}^{\frac{3}{4}} \frac{\sin\varphi}{\cos^3\varphi} d\varphi$$

$$= 128\pi$$

$$= \int_{\frac{\pi}{4}}^{\frac{3}{4}} \frac{1}{\cos^3\varphi} d\cos\varphi$$

$$= \frac{1}{2} \frac{1}{\cos^2\varphi} \Big|_{\frac{\pi}{4}}^{\frac{3}{4}} = \frac{1}{2} \times (2 - 2) = 1$$

e.g.3 $f(x) > 0$, 连续, $F(t) = \frac{\iiint_{\Omega(t)} f(x^2+y^2+z^2) dv}{\iint_{D(t)} f(x^2+y^2) d\sigma}$, $G(t) = \frac{\int_{-t}^t f(x^2+y^2) dx}{\int_{-t}^t f(x^2) dx}$

$\Omega(t): x^2+y^2+z^2 \leq t^2$, $D(t): x^2+y^2 \leq t^2$, 求证: $F(t) \geq \frac{2}{\pi} G(t)$

$$F(t) = \frac{\int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^t f(r^2) r^2 \sin\varphi dr}{\dots\dots\dots} = \frac{4\pi \int_0^t r^2 f(r^2) dr}{2\pi \int_0^t r f(r^2) dr}$$

$$\int_{-t}^t f(x^2) dx = 2 \int_0^t f(r^2) dr$$

$$= 2 \frac{\int_0^t r^2 f(r^2) dr}{\int_0^t r f(r^2) dr}$$

$$G(t) = 2 \cdot \frac{\int_0^t r f(r^2) dr}{\int_0^t f(r^2) dr}$$

$$F(t) \geq \frac{2}{\pi} G(t) \Rightarrow \int_0^t r^2 f(r^2) dr \cdot \int_0^t f(r^2) dr \geq \left[\int_0^t r f(r^2) dr \right]^2$$

柯西-施瓦茨不等式:

$$\left[\int_a^b f(x)g(x) dx \right]^2 \leq \int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx$$

轮换对称性: D : 关于 $y=x$ 对称

$$\iint_D f(x,y) d\sigma = \iint_D f(y,x) d\sigma = \frac{1}{2} \iint_D (f(x,y) + f(y,x)) d\sigma$$

$$\int_0^t r^2 f(r^2) dr \cdot \int_0^t f(s^2) ds - \int_0^t r f(r^2) dr \cdot \int_0^t s f(s^2) ds \geq 0$$

$$\iint_D r^2 f(r^2) f(s^2) dr ds - \iint_D r f(r^2) \cdot s f(s^2) dr ds$$

$D: 0 \leq r \leq t, 0 \leq s \leq t \Rightarrow$ 关于 $r=s$ 对称

$$= \frac{1}{2} \iint_D [r^2 f(r^2) f(s^2) + s^2 f(s^2) f(r^2)] dr ds$$

$$- \frac{1}{2} \iint_D [r f(r^2) f(s^2) \cdot s + s \cdot f(s^2) \cdot r f(r^2)] dr ds$$

$$= \frac{1}{2} \iint_D f(r^2) \cdot f(s^2) [r^2 + s^2 - 2rs] dr ds$$

$$= \frac{1}{2} \iint_D [f(r^2) \cdot f(s^2) (r-s)^2] dr ds \geq 0$$

3. 对称性

A. 奇偶对称性

关于 $xOy / \dots / \dots$ 对称 $f(x, y, z) = -f(x, y, -z)$

引例: $\Omega: x^2 + y^2 + z^2 \leq 1$

$$\iiint_{\Omega} \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dV \Rightarrow I = 0$$

B. 轮换对称性

若 x, y, z 中两个变量互换位置积分区域形式不变,

则被积函数作相同轮换, 三重积分值不变

e.g. $x^2 + y^2 + 2z^2 = 1$ $x \leftrightarrow y$ 不变. $\iiint f(x, y, z) = \iiint f(y, x, z)$

e.g.2 $\Omega_1: x^2 + y^2 + z^2 \leq R^2, \underline{z \geq 0}$ (关于 xOz, yOz 对称)

$\Omega_2: x^2 + y^2 + z^2 \leq R^2, \underline{z \geq 0, y \geq 0, x \geq 0}$ 不对称

A. $\iiint_{\Omega_1} \textcircled{x} dV = 4 \int_0^R \int_0^{\sqrt{R^2 - z^2}} \int_0^{\sqrt{R^2 - z^2 - y^2}} x dx dy dz$ $x: \text{奇} = 0$

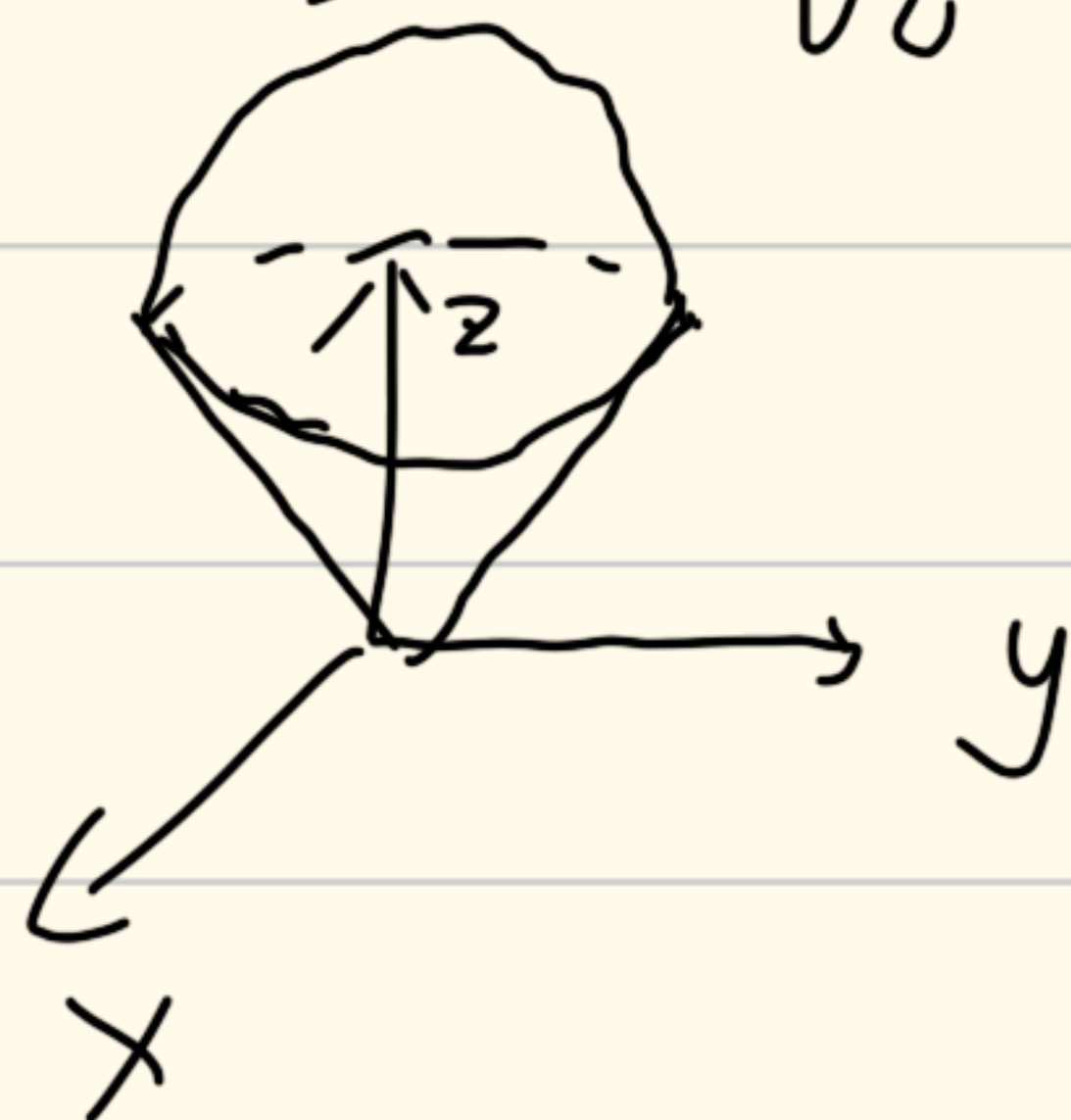
$$B, \int_1^{\overset{xOz}{y}} dv = 4 \int_2^y y dv = 0$$

$$C, \int z dv = 4 \int_2^z z dv \quad \checkmark$$

$$D, \int_1^x xyz dv = 4 \int_2^x xyz dv = 0$$

e.g. $\iiint_{\Omega} (y+z) dv,$

$$\begin{cases} z = \sqrt{x^2+y^2} \\ z = \sqrt{1-x^2-y^2} \end{cases}$$



Ω : 关于 xOz , yOz 对称

$$\Rightarrow \iiint_{\Omega} z dv.$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \varphi \leq \frac{\pi}{4}$$

$$0 \leq r \leq 1$$

$$\begin{aligned} I &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 r \cos \varphi r^2 \sin \varphi dr \\ &= 2\pi \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{\pi}{8} \end{aligned}$$

e.g. 4. $I = \iiint (x^2 + y^2) dV, \quad x \geq 0, y \geq 0, z \geq 0, x^2 + y^2 + z^2 \leq 4$

$xy/yz/xz$ 换位后, 区域不变 $\Rightarrow$ 被积函数中作相同变换结果不变

$$I = \frac{2}{3} \iiint (x^2 + y^2 + z^2) dV.$$

$$= \frac{2}{3} \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^2 r^4 \sin \varphi \cdot dr.$$

$$= \frac{2}{3} \cdot \frac{\pi}{2} \cdot \frac{32}{5} = \frac{32}{15} \pi$$

$$\frac{1}{5} r^5 \Big|_0^2$$

$$- \cos \varphi \Big|_0^{\frac{\pi}{2}}$$

e.g. 5. $V: x^2 + y^2 + z^2 \leq 1, \quad \text{证: } \iiint_V (ax + by)^2 dV = (a^2 + b^2) \iiint_V z^2 dV \quad (a^2 + b^2 \neq 0)$

$$\iiint_V (a^2 x^2 + b^2 y^2 + \underbrace{2abxy}_{=0}) dV = \iiint_V (a^2 x^2 + b^2 y^2) dV.$$

轮换对称

$$\iiint x^2 dV = \iiint y^2 dV = \iiint z^2 dV$$

计算: $I = (a^2 + b^2) \iiint_V z^2 dV.$

法一：球坐标系

$-r, \varphi$

$$= \frac{a^2+b^2}{3} \iiint (x^2+y^2+z^2) dv.$$

$$= \int_0^{2\pi} \int_0^{2\pi} d\theta \int_0^2 d\varphi \int_0^1 r^2 \cdot r^2 \sin \varphi dr \cdot \frac{a^2+b^2}{3}$$

$$= 2\pi \cdot 2 \cdot \frac{1}{3} \cdot \frac{4}{5} \cdot \frac{a^2+b^2}{3}$$

$$= \frac{4}{15} (a^2+b^2) 2$$

法二：柱坐标

$$\iiint_V z^2 dv, \dots$$

$\rightarrow$ 关于 z 偶函数.

V : 关于 x, y 面对称

法三：直角

只关于一个变量有关, 因此用截面法

重积分的换元法

A. 二重积分的换元法 —— 以广义极坐标为例

$$dx dy = |J| dr d\theta$$

$$\begin{vmatrix} x_r' & x_\theta' \\ y_r' & y_\theta' \end{vmatrix}$$

B. 三重积分换元法

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} \begin{matrix} \rightarrow \text{行} \\ \downarrow \text{列} \end{matrix}$$

广义球坐标

$$x = ar \cos \theta \sin \varphi \quad y = br \sin \theta \sin \varphi \quad z = cr \cos \varphi$$
$$|J| = abc r^2 \sin \varphi$$

几何意义

1. 求面积/体积

$$\text{曲面面积公式} \quad \begin{cases} S = \iint_D \sqrt{1 + f_x^2(x,y) + f_y^2(x,y)} d\sigma \\ S = \iint_{D'} \sqrt{EG - F^2} du dv. \end{cases}$$

$$\vec{u} = (x_u, y_u, z_u) \quad \vec{v} = (x_v, y_v, z_v)$$

$$\begin{cases} E = x_u^2 + y_u^2 + z_u^2 \\ F = x_u x_v + y_u y_v + z_u z_v \\ G = x_v^2 + y_v^2 + z_v^2 \end{cases}$$

中值定理:(连续)

$$S = \iint_{D'} \sqrt{|\vec{u}|^2 |\vec{v}|^2 - (\vec{u} \cdot \vec{v})^2} du dv.$$

$$\iint_D f(x,y) d\sigma = f(\xi, \eta) \cdot S_D$$

$$\iiint_{V_n} f(x,y,z) dv = f(\xi, \eta, \theta) \cdot V_n$$

e.g.1

$$\lim_{\rho \rightarrow 0} \frac{1}{2\rho^2} \iint_{x^2+y^2 \leq \rho^2} f(x,y) d\sigma$$

$$\iint_{x^2+y^2 \leq \rho^2} f(x,y) d\sigma = f(\xi, \eta) \cdot 2\rho^2, \exists (\xi, \eta) \in x^2+y^2 \leq \rho^2$$

$$\Rightarrow \lim_{\rho \rightarrow 0} f(\xi, \eta) \Rightarrow \lim_{(\xi, \eta) \rightarrow (0,0)} f(\xi, \eta) = f(0,0)$$

补充结论 ① 由射线 $\theta = \alpha$, $\theta = \beta$, ^{曲线} $r = r(\theta)$ ($\alpha \leq \theta \leq \beta$) 所围区域 D 的面积为

$$\frac{1}{2} \int_{\alpha}^{\beta} [r(\theta)]^2 d\theta$$